

# **DOT NET CORE – INTRODUCTION**

## ***1. Explain ASP.NET Core***

ASP.NET Core is an open-source, cross-platform web framework developed by Microsoft. It allows developers to build modern, high-performance web applications, APIs, and cloud-based services. It's built on top of .NET Core and supports hosting on Windows, Linux, and macOS.

## ***2. Explain the key differences between ASP.NET Core and ASP.NET***

ASP.NET Core is cross-platform, open-source, and lightweight, while ASP.NET (Framework) is Windows-only and not fully open-source. ASP.NET Core offers higher performance, built-in dependency injection, and modular middleware architecture.

## ***3. What are the main components of .NET Core***

The main components are:

- CoreCLR – The runtime that executes applications.
- CoreFX – The base class library providing essential APIs.
- CLI (Command-Line Interface) – For building, running, and publishing apps.
- Roslyn Compiler – Used for compiling C# and VB code.

## ***4. What are some characteristics of .NET Core?***

- Cross-platform (Windows, Linux, macOS).
- Open-source and community-driven.
- High performance and lightweight.
- Unified platform for cloud, web, and mobile.
- Side-by-side versioning support.

## ***5. Latest version of .NET Core?***

The latest version as of October 2025 is .NET 9.0.9 (also known as .NET 9). This is the modern, unified version of .NET Core.

## ***6. How is .NET Core SDK different from .NET Core Runtime?***

- .NET Core Runtime: Used to run applications. It contains the CLR and libraries needed to execute apps.
- .NET Core SDK: Includes the runtime plus development tools like compilers, libraries, and CLI tools for building applications.

## **7. Explain Docker in .NET Core.**

Docker is a containerization platform that allows .NET Core applications to run in isolated, lightweight containers. Developers can package the app and its dependencies together for consistent deployment across environments.

## **8. Share specific features of .NET Core.**

- Cross-platform development : Runs on Windows, Linux, and macOS, allowing one codebase for all systems.
- Unified programming model : Single framework for web, desktop, cloud, and mobile apps.
- High performance : Optimized runtime ensures fast execution and low memory use.
- Built-in dependency injection : Improves modularity and simplifies code testing.
- Cloud-ready architecture : Easily integrates with cloud platforms like Azure and AWS.
- Flexible deployment options : Supports both shared and self-contained deployments.
- Open-source with active community support : Free, open-source, and regularly improved by Microsoft and the community.